

KELSON VINCEN

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Research Interests: Spatial database systems and learned index structures for efficient geospatial querying.

EDUCATION

Bina Nusantara University (BINUS)

2023 – Present

Bachelor of Science in Computer Science | Tangerang, Indonesia | Expected Graduation: August 2027

- Cumulative GPA: 3.99 / 4.00
- Dean's List 2025, School of Computer Science
- Relevant Coursework: Database Systems, Big Data, Cloud Computing, Data Structures and Algorithms

PUBLICATIONS

Vincien, K., Harison, A. T., Chandra, A. J., Meiliana, M., and Riftiarrasyid, M. F. (2025). Evaluating Geometric Transformations, FA-GAN Augmentation, and CutMix to Improve Facial Expression Recognition. *2025 7th International Conference on Cybernetics and Intelligent System (ICORIS 2025)*. IEEE. <https://doi.org/10.1109/ICORIS67789.2025.11296013>

[In Preparation] **Vincien, K.**, and Lim, J. A. Balancing User Coverage and Grid Efficiency in EV Charging Station Placement: A Scenario-Based MCLP Framework for Jakarta, Indonesia. *Discover Cities* (Springer Nature). Target submission: July 2026.

RESEARCH EXPERIENCE

Undergraduate Thesis Research

2025 – Present

Bina Nusantara University | Supervisor: Dr. Ir. Edy Irwansyah, S.T., M.Si., IPM, ASEAN Eng.

- Developing a new placement method for a GIS-based facility location model (MCLP) that balances two goals at once, covering as many users as possible and keeping the power grid efficient. The method also measures how unevenly the environmental burden falls across Jakarta's grid substations, something no earlier version of this model has done.
- Building a full spatial modeling pipeline in Python. Manuscript targeting submission to *Discover Cities* (Springer Nature) in July 2026.

Learned Spatial Index Systems

2026 – Present

Research Proposal in Progress

- Identified an open problem. No existing 2D learned spatial index can accept new data without a full rebuild. This update problem is already solved for 1D learned indexes, but not for 2D.
- Conducted a thorough review of foundational work in the field, including RSMI (VLDB 2020), ELSI (ICDE 2023), and Chameleon (ICDE 2024). Proposing a gapped rank space construction that reserves room in the rank assignment for future insertions, so existing points' ranks and curve values stay fixed. The open question is how far the required reorganization spreads once a region's reserved gaps run out.

PROFESSIONAL EXPERIENCE

Data Engineer Intern

Feb 2026 – Present

PT Cyberindo Aditama (CBN) — Technology Development Division | Jakarta, Indonesia

- Maintaining 4 automated data pipelines that run daily across all 13 branches, covering customer records, network traffic logs, a churn-prediction model, and communication analytics.
- Designed an AI agent-driven data migration system that maps and migrates production data from 13+ branch-specific schemas into a single target PostgreSQL schema, including a confidence-scored mapping protocol, automated multi-layer data validation, and a cross-branch key-collision policy. Cut migration time per module from 1-2 months to 1-2 weeks.
- Set up GitLab version control for all NiFi data pipelines, replacing an older tool, NiFi Registry Standalone, that Apache has deprecated. This gives the team a full history of changes, the ability to undo a bad change instantly, and separate environments for testing, staging, and production.
- Conducted an OpenSearch proof-of-concept for a new coverage search engine and built a deduplication automation layer for the CDN pipeline.

LEADERSHIP

Manager, IT Development Division

Feb 2025 – Feb 2026

Binus Student Learning Community | Bina Nusantara University

- Led a team of 10 developers in building and maintaining web platforms that support university learning initiatives, overseeing technical architecture and project execution.
- Contributed to more than 5 organizational events across divisions including Events, Marketing, and Technology, supporting cross-functional coordination and execution.

COMMUNITY ENGAGEMENT

Scholarship Mentor

Sep 2024 – Jan 2025

Bina Nusantara University

- Mentored students in Artificial Intelligence, Algorithm Design and Analysis, Object-Oriented Programming, Computational Physics, and Database Technology.
- Founded an open Discord community for peer study, which grew to over 100 students. Led public review sessions before midterms and finals in Computational Physics and Algorithm Design and Analysis.

Programming Teacher

Sep 2025 – Jan 2026

Timedoor Academy

- Taught programming fundamentals using Python, Scratch, and Roblox to students aged 5 to 18 in small-group settings of 4 to 5 students, adapting instruction across a wide range of ages and skill levels.

Community Teaching Volunteer

2024

East Jakarta, Indonesia

- Taught mathematics and science to elementary school students over a one-week community service program, working in small groups of 6 students across 5 days.

TECHNICAL SKILLS

Python, SQL, QGIS, Apache NiFi, PostgreSQL, SQL Server, GitLab, Git, OpenSearch

AWARDS AND RECOGNITION

- BINUS Appreciation Day (2026): University-wide recognition for outstanding competition achievement.
Awarded by Bina Nusantara University in recognition of the 3rd place finish at the MAPID WebGIS Competition.
- 3rd Place, MAPID WebGIS Competition (2025): GeoVest, a geospatial AI tool for investment property decision-making.
National-level competition hosted by MAPID across Indonesian universities.
- Top 10 Grand Finalist, BINUS x Microsoft Hackathon (2024): EyeFriend, an AI-powered tool for the visually impaired.
Top 10 finish among competing university teams in a hackathon jointly organised by Bina Nusantara University and Microsoft.